



CONVECTION CURRENTS – COLOUR IN WATER



A Fun Science Experiment for You!

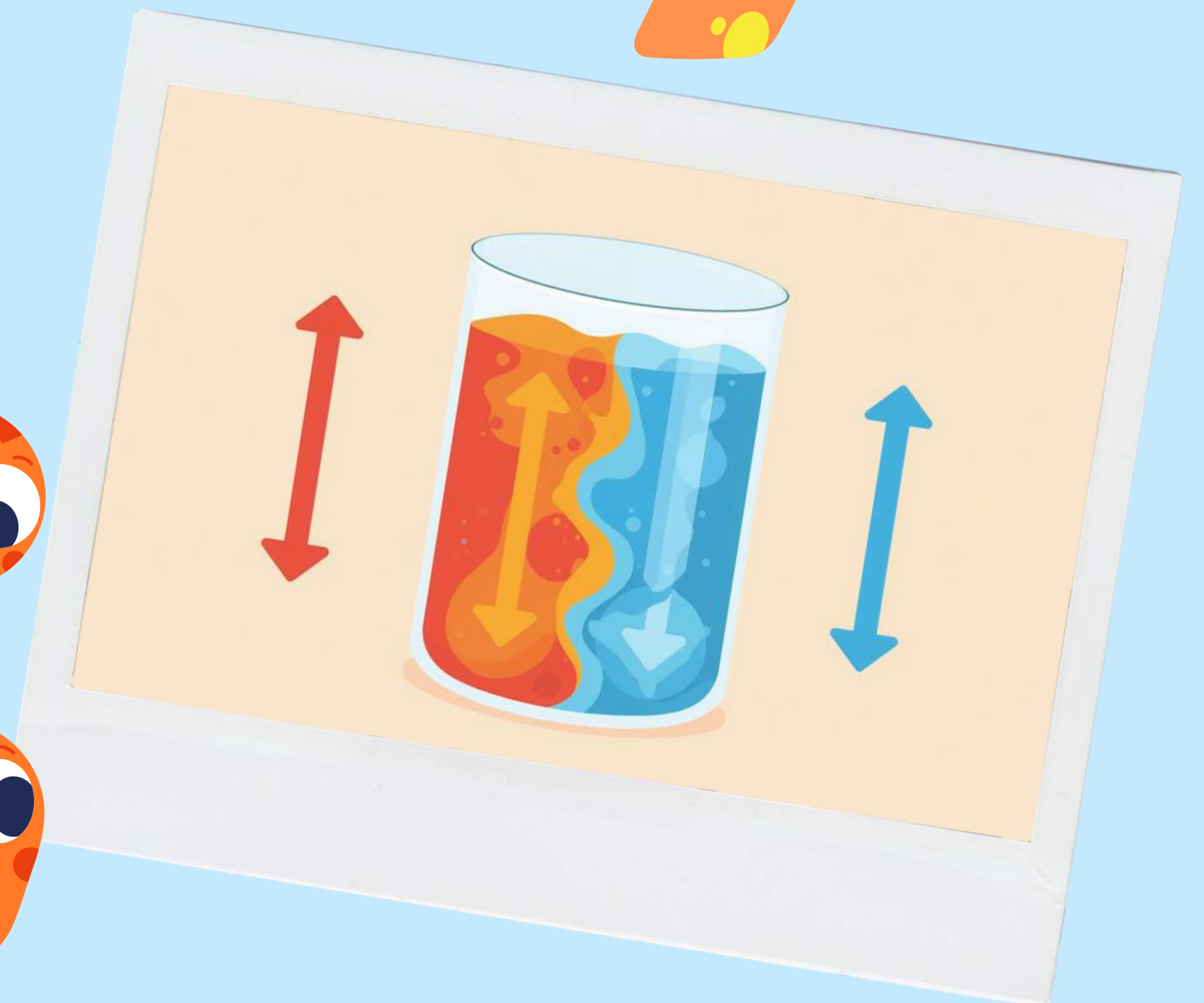
WATCH THE MAGIC!

Kabir drops turmeric in hot water.
Look! The yellow colour swirls up.



WHAT HAPPENS WHEN WATER GETS HOT?

Hot water rises up. Cold water sinks down.
When water gets warm, it becomes lighter and floats to the top. When water is cold, it becomes heavier and moves to the bottom. This is how heat makes water dance!



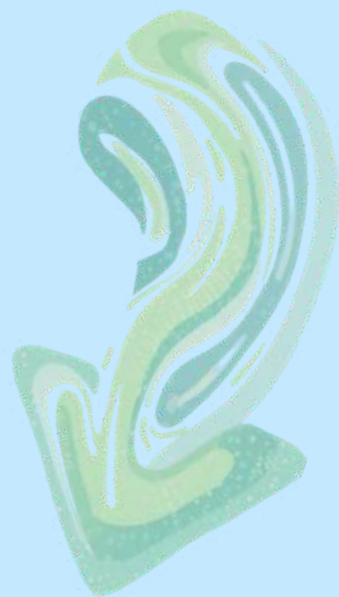


LET'S SEE THE EXPERIMENT!



1

Fill a clear glass with cold water. Use blue colour to show it's cold!



2

Add warm water with yellow or orange colour. Watch it carefully!



3

See the colour move and swirl! The hot water rises up.



WHAT DO YOU SEE?

Hot coloured water rises and swirls at the top!
Watch the yellow and orange colours dance upward.

Cold coloured water sinks to the bottom when added to warm water. The blue colour goes down.

This is how heat makes water move in beautiful patterns!





LET'S PREDICT & PLACE!

Where will the colour go? Hot water or cold water?

Look at the two containers. One has hot water, one has cold water.

Guess before we try! Point to where you think the colour will go.

Teachers & Parents: Let children make predictions before adding colour drops to each container.



HOT RISES, COLD SINKS - OUR GOLDEN RULE

1

Hot water and air always go UP — like steam rising from hot roti!

2

Cold water and air always go DOWN — they sink to the bottom!

3

This up-and-down movement creates a loop called a convection current!

WHAT IS A CONVECTION CURRENT?

Convection current is a loop. Hot parts move UP. Cool parts move DOWN. This keeps going!
Imagine warm water in a pot. The hot water at the bottom rises up. The cooler water at the top sinks down. Round and round it goes, like a merry-go-round! This circular movement is called a convection current. It happens in water, air, and even in the Earth!





CONVECTION VS CONDUCTION

Conduction: Heat moves by TOUCHING
Example: A metal spoon gets hot in dal!

Convection: Heat moves by MOVING
Example: Hot water rises up in a pot!
Remember: Touch = Conduction, Move =

Convection

INDIAN EXAMPLES OF CONVECTION CURRENTS



Sea Breeze in
Mumbai



Sambar on Slow Flame



Agarbatti Smoke Rising



mini QUIZ TIME!

1

What happens to hot water?

Answer: It rises up!

2

What do we call the circular movement of hot and cold water?

Answer: Convection current!

3

What is heat?

Answer: Energy that makes things warm!

LET'S RECAP WHAT WE LEARNED

1. Heat makes water move in a loop called convection current
2. Hot water rises, cold water sinks
3. Colour in water shows this movement
4. Convection is different from conduction
5. We see convection currents in daily life in India



SAFETY FIRST!

Never touch hot water without a grown-up's help!

Always ask an adult to pour or carry hot water. Stay safe while having fun with science experiments!





THANK YOU!
KEEP EXPLORING!

